

Basic User Interface Design for Electronics Engineers



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An engineer studies electronics at the bachelor level, and what he or she learns are the fundamentals of electronics and how to use these for designing circuits. However, when the engineer joins a company, most of the times, he or she gets to work on real-life products, which are completely different from the circuits that are designed in college, as full product design is a completely intense process where users of the products take the primary position in defining product features.

Structured development process

A typical product design has well-defined steps, and the first phase is called concept phase. Large and experienced product companies follow a structured development process. One of the key things that structured product development demands is usability study, which helps product designers to define the shape and size of the product, along with its user interface (UI). The depth of study varies from company to company.

However, if the product is for use in medical, avionics and other safety-critical applications, usability study is critical and needs to be carried out in detail, covering all possibilities to ensure that the UI does not cause wrong usage of the product or introduce errors. Since usability engineering or UI design (UID) is a specialised area, it often requires special training. Engineers who are trained in UID are called industrial design (ID) engineers. They are trained in aspects like usability, ergonomics and visual cues, among others.

Large companies have specially-trained designers for UID. Smaller companies find it difficult to hire their services as these are expensive. This article aims to help designers who work in smaller companies, as well as professional electronics designers, to design products with essential UI features

in their designs. A typical UID has two essential parts:

1. Basic UI rules, which are mandatory in most products
2. Special needs depending on product categories like consumer, safety critical, medical and avionics, which need professional help in designing the UI

What we will see here are standard UID inputs, which will help designers meet the bare minimum UI needs. While this is not exhaustive content, it is good enough to meet the needs of small products, which have no critical-safety functionality requirements.

Usability study

Usability study is a process where ID engineers create product mock-ups (in most cases) and test these with prospective users for feedback. Typically, they create four to five variations to get a feel of what the end customer wants. Spreading the features across three to four mock-ups, instead of putting all features in one, ensures that users will not get overwhelmed and shall provide the feedback in an objective way. This type of study reveals some critical aspects of the product, such as:

1. Shape and size of the product (especially, if the product is handheld or portable)
2. A keyboard, its functions and layout
3. An output devices like a display or an alarm, and its relationship to an input device like a keyboard, knob or lever
4. Sequence of product functions and how users interact with the product

In all these, the focus is on UI so that the user does not commit any induced errors.

Industrial designers are trained to use the mock-up or, sometimes, the functional prototype of the product itself to elicit feedback from prospective users and use the